8.6

(a) $H_0: \sigma_M^2 = \sigma_F^2$ vs $H_1: \sigma_M^2 \neq \sigma_F^2$

$N_M = 577$, $N_F = 1000 - 577 = 423$

$df_M = 573$, $df_F = 419$

$\hat{\sigma}_M^2 = \frac{SSE_M}{573} = \frac{97161.9174}{573} = 169.567$

$\hat{\sigma}_F^2 = (12.024)^2 = 144.577$

$F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.567}{144.577} = 1.173 \sim F_{573,419}$

Rejection region: $F < F_{0.025,573,419} = 0.838$ or

$F > F_{0.975,573,419} = 1.197$

Because $0.838 < F = 1.173 < 1.197$, do not reject H_0.

(b) $H_0: \sigma^2_{SINGLE} = \sigma^2_{MARRIED}$ vs $H_1: \sigma^2_{MARRIED} > \sigma^2_{SINGLE}$

$df_s = 400 - 5 = 395$, $df_m = 600 - 5 = 595$

$\hat{\sigma}_s^2 = \frac{56231.0382}{395} = 142.357$

$\hat{\sigma}_M^2 = \frac{100703.0471}{595} = 169.249$

$F = \frac{169.249}{142.357} = 1.189$

Rejection region: $F > F_{0.95,595,395} = 1.165$

Because $F = 1.189 > 1.165$, we reject H_0.

(c) H_0: homoskedasticity vs H_1: heteroskedasticity

$NR^2 \sim \chi^2(4)$

$\chi^2_{0.05}(4) = 9.488 < NR^2 = 59.03$

$\therefore$ Reject H_0. We conclude that there is strong evidence of heteroskedasticity. This evidence provide no evidence about the error variation is different for married and unmarried individuals.

(d) 原变数 (4): EDUC, EXPER, METRO, FEMALE

平方项 (2): $EDUC^2$, $EXPER^2$

交乘项 (6): $\binom{4}{2}$

$df = 4 + 2 + 6 = 12$

$\chi^2_{0.05}(12) = 21.026$

$\because 21.026 < 78.82$, reject H_0. We conclude that there is strong evidence of heteroskedasticity.

```
> qf(0.025, 573, 419)
[1] 0.837669
> qf(0.975, 573, 419)
[1] 1.196781
> qf(0.95, 595, 395)
[1] 1.164705
> qchisq(p = 0.95, df = 4)
[1] 9.487729
> qchisq(p = 0.95, df = 12)
[1] 21.02607
```

(e) narrower : EXPER、METRO、FEMALE

wider : Intercept、EDUC

No inconsistency. Robust standard error 不一定比較大. 存在異質變異時, usual SE 可能高估, 也可能低估.

(f) It is compatible. (b) 是檢定婚姻狀態是否影響誤差變異.

(f) 是檢定婚姻狀態是否影響平均薪資.

Since $t = 1 < 1.96$, it indicates that marital status has no significant effect on the average wage.

一个變數可以影响誤差變異, 但不影响平均數.

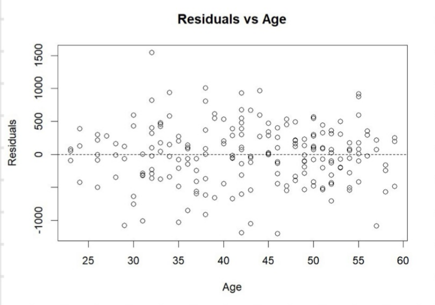
8.16

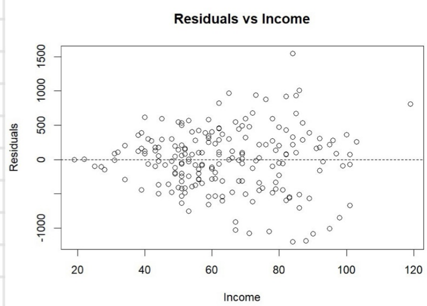
(a)

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation)

Residuals:
     Min       1Q   Median       3Q      Max
-1198.14  -295.31    17.98   287.54  1549.41

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -391.548    169.775  -2.306   0.0221 *
income        14.201      1.800   7.889 2.10e-13 ***
age           15.741      3.757   4.189 4.23e-05 ***
kids         -81.826     27.130  -3.016   0.0029 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared:  0.3406,	Adjusted R-squared:  0.3305
F-statistic: 33.75 on 3 and 196 DF,  p-value: < 2.2e-16

> confint(mod, "kids", level = 0.95)
         2.5 %    97.5 %
kids -135.3298 -28.32302
```

Holding household income and average adult age constant, for each additional child decrease the expected miles traveled by 81.8 miles.

(b)





- Residual vs Income : funnel shape, the spread of the residuals widens as income increase. This pattern strongly suggests the presence of heteroskedasticity.
- Residual vs age : randomly dispersed

(c)

```
> Fstat
[1] 3.104061
> qf(0.95, 86, 86)
[1] 1.428617
```

$H_0: \sigma^2_{low} = \sigma^2_{high}$ vs $H_1: \sigma^2_{High} > \sigma^2_{low}$

$F = 3.104 > F_{0.05,86,86} = 1.429$. Reject H_0, it is enough evidence to conclude that the error variance is larger for the high income group and is related to the magnitude of income.

(d)

```
> coeftest(mod, vcov = vcovHC(mod, type = "HC1"))

t test of coefficients:

             Estimate Std. Error t value  Pr(>|t|)
(Intercept) -391.5480   142.6548 -2.7447 0.0066190 **
income        14.2013     1.9389  7.3246 6.083e-12 ***
age           15.7409     3.9657  3.9692 0.0001011 ***
kids         -81.8264    29.1544 -2.8067 0.0055112 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

> beta_kids <- coef(mod)["kids"]
> se_kids_r <- sqrt(vcovHC(mod, type = "HC1")["kids","kids"])
> beta_kids + c(-1,1)*qt(0.975, df=196)*se_kids_r
[1] -139.32297  -24.32986
```

(a) : $(-135.33, -28.32)$

(d) : $(-139.32, -24.33)$

Robust interval estimate becomes wider than the conventional interval in (a).

(e)

```
> confint(mod_gls, "kids", level = 0.95)
         2.5 %    97.5 %
kids -119.8945 -33.71808
>
> coeftest(mod_gls, vcov = vcovHC(mod_gls, type = "HC1"))

t test of coefficients:

             Estimate Std. Error t value  Pr(>|t|)
(Intercept) -424.9962    95.8035 -4.4361 1.526e-05 ***
income        13.9473     1.3470 10.3545 < 2.2e-16 ***
age           16.7175     2.7974  5.9761 1.061e-08 ***
kids         -76.8063    22.6186 -3.3957 0.0008286 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

> beta_kids_gls <- coef(mod_gls)["kids"]
> se_kids_gls_r <- sqrt(vcovHC(mod_gls, type = "HC1")["kids","kids"])
> beta_kids_gls + c(-1,1)*qt(0.975, df=196)*se_kids_gls_r
[1] -121.41339  -32.19919
```

OLS usual : $(-135.33, -28.32)$, OLS robust : $(-139.32, -24.33)$

GLS usual : $(-119.89, -33.72)$, GLS robust : $(-121.41, -32.20)$

The GLS interval is narrower than the OLS interval, indicating that GLS is more efficient under the assumption that $\sigma_i^2 = \sigma^2 INCOME_i^2$.